Lecture 8

Application II: selection rules and level splitting

This lecture decides which of ammonia's vibrations absorb infrared light, and light of which polarisation. Some transitions are forbidden by symmetry alone, whatever the forces in the molecule. An operator can connect two levels only if the final level occurs in the tensor product of the operator and the initial level: that is the selection rule, and the Wigner–Eckart theorem sharpens it. The rule gives the infrared and Raman rules. Finally, a perturbation that lowers the symmetry splits the levels, and restricting characters to the smaller group tells how.

By the end of this lecture you will be able to:

- Decompose tensor products by characters.
- Apply the selection rule and the Wigner–Eckart theorem.
- Decide infrared and Raman activity, and how levels split when symmetry is lowered.

8.1 Tensor products and Clebsch–Gordan decomposition

If operators T_α transform by matrices $D_T(g)$ and states $|i_\nu\rangle$ by $D^{(i)}(g)$, the vectors $T_\alpha\,|i_\nu\rangle$ transform by the Kronecker products $D_T(g)\otimes D^{(i)}(g)$ (Section 8.2). Decomposing such products decides which levels a transition can reach.

Recall the tensor product of vector spaces from linear algebra. Let V, W be vector spaces of dimensions n and m, with bases (e_μ), (f_ν). Their *tensor product* $V\otimes W$ is the vector space with basis the nm symbols $e_\mu\otimes f_\nu$. For general vectors, $v\otimes w$ is defined by bilinear expansion, so its coordinates are the Kronecker product of the coordinates of v and w. Linear maps X on V and Y on W give the map $X\otimes Y$ on $V\otimes W$, $(X\otimes Y)(v\otimes w)=Xv\otimes Yw$, whose matrix is the Kronecker product of their matrices. (Up to canonical isomorphism, the construction does not depend on the bases.)

Definition 8.1 (Tensor product of representations)**.** Let (ρ_V, V) and (ρ_W, W) be representations of the same group G. Their *tensor product representation* on $V\otimes W$ acts factorwise:

$$(\rho_V\otimes\rho_W)(g)\,(v\otimes w)=\rho_V(g)\,v\ \otimes\ \rho_W(g)\,w,$$

extended linearly.

If ρ_V and ρ_W have matrices $D_V(g)$ and $D_W(g)$, the matrix of $(\rho_V\otimes\rho_W)(g)$ is the Kronecker product $D_V(g)\otimes D_W(g)$. The mixed-product rule of Lecture 6 gives the homomorphism property: $\big(D_V(g)\otimes D_W(g)\big)\big(D_V(h)\otimes D_W(h)\big)=D_V(gh)\otimes D_W(gh)$.

Proposition 8.2 (Characters multiply). $\chi_{V\otimes W} = \chi_V \cdot \chi_W$.

Proof. The trace of a Kronecker product is the product of the traces: $\operatorname{tr}(X\otimes Y) = \sum_{\mu,\alpha} X_{\mu\mu}Y_{\alpha\alpha} = \operatorname{tr}X \operatorname{tr}Y$. ∎

Products of irreducible representations are generally reducible, and characters decompose them at once.

Definition 8.3 (Clebsch–Gordan series and coefficients). The decomposition

$$V^{(i)} \otimes V^{(j)} \;\cong\; \bigoplus_k c_{ij}^{k}\, V^{(k)}, \qquad c_{ij}^{k} = \left\langle \chi_k,\, \chi_i\chi_j \right\rangle_G ,$$

is the *Clebsch–Gordan series*. Pass from the orthonormal product basis $e_\mu^{(i)} \otimes e_\nu^{(j)}$ to an orthonormal symmetry-adapted basis of the right-hand side by a unitary change of basis. The entries of this change of basis are the *Clebsch–Gordan coefficients* $\langle k, \kappa, (s) \mid i, \mu; j, \nu\rangle$. Here κ numbers the partners of $V^{(k)}$, and s numbers the c_{ij}^k copies when the multiplicity exceeds one.

Example 8.4 (Clebsch–Gordan table of C_{3v}). We use the character table of Lecture 5. A_1 is the identity of the product, $A_2\otimes A_2 = A_1$, and $A_2\otimes E$ has character $(1,1,-1)\cdot(2,-1,0) = (2,-1,0)$, so $A_2 \otimes E = E$. The interesting entry is $\chi_E^2 = (4,1,0)$:

$$m_{A_1} = \tfrac{1}{6}(4+2) = 1, \quad m_{A_2} = \tfrac{1}{6}(4+2) = 1, \quad m_E = \tfrac{1}{6}(8-2) = 1,$$

$$E \otimes E = A_1 \oplus A_2 \oplus E.$$

The decomposition can be seen directly. Let $\vec a$ and $\vec b$ be two vectors in the plane of the triangle, each transforming by the defining representation E. The four components $a_\alpha b_\beta$ of $\vec a \otimes \vec b$ fall into three families. The dot product $a_xb_x + a_yb_y$ is unchanged by every rotation and reflection: A_1. The cross product $a_xb_y - a_yb_x$ is unchanged by the rotations and changes sign under a reflection: A_2. The remaining pair $(a_xb_x - a_yb_y,\; a_xb_y + a_yb_x)$ turns through 2θ when the plane turns through θ, so a rotation through $2\pi/3$ acts on it as a rotation through $4\pi/3$, with trace $2\cos\frac{4\pi}{3} = -1$, and a reflection has trace 0 on it: E again. Normalised, the coefficients of these combinations are Clebsch–Gordan coefficients. For instance, $(a_xb_x + a_yb_y)/\sqrt{2}$ gives $\langle A_1 \mid E, \mu; E, \nu\rangle = \delta_{\mu\nu}/\sqrt{2}$. ◁

One refinement of the character calculus will be needed for Raman scattering.

Proposition 8.5 (Symmetric and antisymmetric squares). $V \otimes V$ splits invariantly into symmetric and antisymmetric parts, $S^2V \oplus \Lambda^2V$, with characters

$$\chi_{S^2V}(g) = \tfrac{1}{2}\left[\chi(g)^2 + \chi(g^2)\right], \qquad \chi_{\Lambda^2V}(g) = \tfrac{1}{2}\left[\chi(g)^2 - \chi(g^2)\right].$$

Proof. Let P be the swap, $P(v\otimes w) = w\otimes v$. It commutes with every $\rho(g)\otimes\rho(g)$, since both products send $v\otimes w$ to $\rho(g)w\otimes\rho(g)v$. So its eigenspaces S^2V ($P=+1$) and Λ^2V ($P=-1$) are invariant, with projectors $\frac{1}{2}(1\pm P)$, and

$$\chi_{S^2V}(g) = \tfrac{1}{2}\left[\operatorname{tr}\big(D_V(g)\otimes D_V(g)\big) + \operatorname{tr}\big(P\,(D_V(g)\otimes D_V(g))\big)\right].$$

The first trace is $\chi(g)^2$. For the second, $P(X\otimes Y)$ sends $e_\mu\otimes e_\nu$ to $Ye_\nu\otimes Xe_\mu$, whose $e_\mu\otimes e_\nu$ coefficient is $Y_{\mu\nu}X_{\nu\mu}$; so $\operatorname{tr}\big(P(X\otimes Y)\big) = \operatorname{tr}(XY)$, and the second trace is $\operatorname{tr}D_V(g)^2 = \chi(g^2)$. The sign $-$ gives Λ^2V. ∎

For E of C_{3v}, swapping $\vec{a}$ and $\vec{b}$ in Example 8.4 fixes the dot product and the pair and reverses the cross product, so $S^2E = A_1 \oplus E$ and $\Lambda^2 E = A_2$. The formula agrees: $\chi_E(g^2) = (2,-1,2)$, and $\frac{1}{2}\big[(4,1,0)+(2,-1,2)\big] = (3,0,1)$.

Pairing with the dual, and invariant tensors *[self-study]*

Proposition 8.6 (Pairing with the dual)**.** Let $V^{(i)}, V^{(j)}$ be irreducible representations. The trivial representation occurs in $V^{(i)} \otimes V^{(j)}$ if and only if $V^{(j)} \cong \big(V^{(i)}\big)^*$, and then exactly once.

Proof. The multiplicity of the trivial representation is

$$\langle \chi_{\text{triv}}, \chi_i \chi_j \rangle_G = \frac{1}{|G|}\sum_g \chi_i(g)\chi_j(g) = \langle \overline{\chi_i}, \chi_j \rangle_G = \langle \chi_{i^*}, \chi_j \rangle_G = \delta_{i^*,j},$$

where we used $\chi_{i^*} = \overline{\chi_i}$ (the self-study section of Lecture 5), the fact that the dual of an irreducible representation is irreducible (so χ_{i^*} is one of the χ_k), and the first orthogonality relation. ∎

Let a vector T in a representation satisfy $\rho(g)T = T$ for all g. Equivalently, T spans a copy of the trivial representation. Such a vector is an *invariant*. An invariant in a tensor power $V \otimes \cdots \otimes V$ (possibly with dual factors) is an *invariant tensor*. The number of invariants is one character average:

$$\dim\{\text{invariants}\} = \langle \chi_{\text{triv}}, \chi \rangle_G = \frac{1}{|G|}\sum_g \chi(g),$$

that is, the multiplicity of the trivial representation.

Take the representation $V = E$ of C_{3v}, with $\chi = (2,-1,0)$ on the classes $(E, 2C_3, 3\sigma_v)$. In $V \otimes V$, the character $\chi^2 = (4,1,0)$ gives $\frac{1}{6}(4+2+0) = 1$ invariant. It is the invariant bilinear form, i.e. the $\delta_{\mu\nu}$ left untouched by the orthogonal matrices: the dot product of Example 8.4. In $V \otimes V \otimes V$, the character $\chi^3 = (8,-1,0)$ gives $\frac{1}{6}(8-2+0) = 1$ invariant cubic. It is a symmetric three-index tensor $d_{\mu\nu\lambda}$ peculiar to the triangle's geometry.

Invariant tensors are the rigid skeleton of a symmetry. Every invariant interaction, coupling or form a theory can contain must be built from them. In Part II we apply this principle to SU(2), and it will dictate which terms may appear in physical Hamiltonians and Lagrangians. For SU(3) the same idea fixes the δ, ϵ and d, f tensors used in strong-interaction physics, but that is further study. This course does not teach that SU(3) application, nor its triplet, antitriplet and eightfold-way content. Jones, Ch. 8, §§8.3–8.4 is the place to read about them.

8.2 Selection rules and the Wigner–Eckart theorem

Let G be a symmetry of a quantum system, acting by the unitary operators $\rho(g)$ as in Lecture 6. Operators, like states, organise into symmetry multiplets: the group acts on them by conjugation, $A \mapsto \rho(g) A \rho(g)^{-1}$.

Definition 8.7 (Irreducible tensor operator)**.** A family of operators T_α, $\alpha = 1, \dots, n_T$,

transforms as the representation V_T, with matrices $D_T(g)$, if

$$\rho(g)\, T_\alpha\, \rho(g)^{-1} = \sum_\beta D_T(g)_{\beta\alpha}\, T_\beta \qquad \text{for all } g \in G.$$

If V_T is irreducible, the family is an *irreducible tensor operator*. The trivial type means $[\rho(g), T] = 0$: an *invariant* (scalar) operator, such as the Hamiltonian itself. The position operators x, y, z of a particle, and so the electric dipole moment that couples it to light, transform like the coordinate functions, by the vector representation $R(g)$.

Let $\mathcal{L}_i$ be an energy level carrying the irreducible $V^{(i)}$, with an orthonormal basis $|i_\nu\rangle$ transforming by unitary matrices $D^{(i)}$, and let $\mathcal{L}_f$ be another level, carrying $V^{(f)}$. (We write $\mathcal{L}$ for a level, since V_i denotes an isotypic component.) The numbers $\langle f_\mu | T_\alpha | i_\nu \rangle$ are the amplitudes for transitions from $\mathcal{L}_i$ to $\mathcal{L}_f$ caused by the T_α.

Theorem 8.8 (Selection rule)**.** The map

$$\Phi : V_T \otimes \mathcal{L}_i \to \mathcal{L}_f, \qquad \Phi(t_\alpha \otimes |i_\nu\rangle) = \pi_f\, T_\alpha\, |i_\nu\rangle\,,$$

where t_α is the basis of V_T and π_f the orthogonal projection onto $\mathcal{L}_f$, is an intertwiner, and its matrix elements are $\langle f_\mu | T_\alpha | i_\nu \rangle$. Consequently all of them vanish unless $V^{(f)}$ occurs in $V_T \otimes V^{(i)}$, that is, unless

$$\langle \chi_f, \chi_T\, \chi_i \rangle_G \neq 0.$$

Proof. Insert $\rho(g)^{-1}\rho(g)$:

$$\rho(g)\, T_\alpha\, |i_\nu\rangle = \big(\rho(g) T_\alpha \rho(g)^{-1}\big)\rho(g)\, |i_\nu\rangle = \sum_{\beta,\lambda} D_T(g)_{\beta\alpha}\, D^{(i)}(g)_{\lambda\nu}\, T_\beta\, |i_\lambda\rangle\,.$$

The right-hand side is the image under $t_\beta \otimes |i_\lambda\rangle \mapsto T_\beta\, |i_\lambda\rangle$ of the product action on $t_\alpha \otimes |i_\nu\rangle$. The projection π_f commutes with every $\rho(g)$, since $\mathcal{L}_f$ and its orthogonal complement are invariant. So $\rho(g)\Phi = \Phi\,(\rho_T \otimes \rho_i)(g)$. The matrix elements of Φ are $\langle f_\mu | \pi_f T_\alpha | i_\nu \rangle = \langle f_\mu | T_\alpha | i_\nu \rangle$. By Lecture 5 the intertwiners from $V_T \otimes V^{(i)}$ to the irreducible $V^{(f)}$ form a space of dimension $\langle \chi_f, \chi_T \chi_i \rangle_G$. If this is 0, then $\Phi = 0$. ∎

The rule is a necessary condition only: symmetry can forbid a transition, but it cannot guarantee one. A transition allowed by symmetry may still have a small amplitude, for reasons the symmetry does not see. In words: *a transition is allowed only if the final representation occurs in the product of the operator's and the initial state's.*

Theorem 8.9 (Wigner–Eckart, multiplicity-free)**.** Suppose $V^{(f)}$ occurs exactly once in $V_T \otimes V^{(i)}$, and fix a nonzero intertwiner $C : V_T \otimes V^{(i)} \to V^{(f)}$, with components

$$C\big(t_\alpha \otimes e^{(i)}_\nu\big) = \sum_\mu C^\mu_{\alpha\nu}\, e^{(f)}_\mu.$$

Then there is a single number $\langle f \| T \| i \rangle$, the *reduced matrix element*, such that

$$\langle f_\mu | T_\alpha | i_\nu \rangle = \langle f \| T \| i \rangle\, C^\mu_{\alpha\nu} \quad \text{for all } \alpha, \mu, \nu.$$

Proof. Identify the levels with $V^{(i)}$ and $V^{(f)}$ through their bases. Then Φ is an intertwiner $V_T \otimes V^{(i)} \to V^{(f)}$, and its components are the matrix elements. These intertwiners form a one-dimensional space, so $\Phi = \langle f\|T\|i\rangle\, C$, and the components agree. ∎

When C is the isometry onto the copy of $V^{(f)}$ in the orthonormal adapted basis of Definition 8.3, the numbers $C^{\mu}_{\alpha\nu}$ are the *Clebsch–Gordan coefficients*; any other choice of C only rescales $\langle f\|T\|i\rangle$. They depend only on the group and the three representations, not on the physics: of the $n_T\, n_i\, n_f$ amplitudes, the physics is in one number. For an invariant operator A, the type V_T is trivial, the coefficients of $A_1 \otimes V^{(i)} \to V^{(i)}$ are $\delta_{\mu\nu}$, and the theorem reads $\langle w_{\mu,a}, A\, w_{\nu,b}\rangle = \delta_{\mu\nu} F^{(i)}_{ab}$. That is the block theorem of Lecture 6, proved there by the same count of intertwiners; the entries of the reduced matrix are the reduced matrix elements. In Part II the same argument, for the rotation group, is the Wigner–Eckart theorem of atomic physics.

[self-study] *With multiplicities.* If $V^{(f)}$ occurs $c > 1$ times in $V_T \otimes V^{(i)}$, the same argument gives c independent intertwiners $C_1, \dots, C_c$, and the amplitude is a sum of c terms, $\langle f_\mu | T_\alpha | i_\nu\rangle = \sum_{s=1}^{c} \langle f\|T\|i\rangle_s\, (C_s)^{\mu}_{\alpha\nu}$: c reduced matrix elements per triple of types.

8.3 Infrared and Raman activity

A molecule absorbs infrared light of frequency ω when its electric dipole moment oscillates at ω, since the light's electric field $\vec{\mathcal{E}}(t)$ couples to the dipole through the energy $-\vec{\mathcal{E}} \cdot \vec{d}$. Let $\vec{d}(q) \in \mathbb{R}^3$ be the dipole moment of the molecule, nuclei and electrons together, when its nuclei are displaced to q (Lecture 7). Carrying the displaced molecule by a symmetry operation g turns its dipole by $R(g)$, and the dipole does not care which of two like nuclei is which, so

$$\vec{d}\big(\rho(g)q\big) = R(g)\, \vec{d}(q).$$

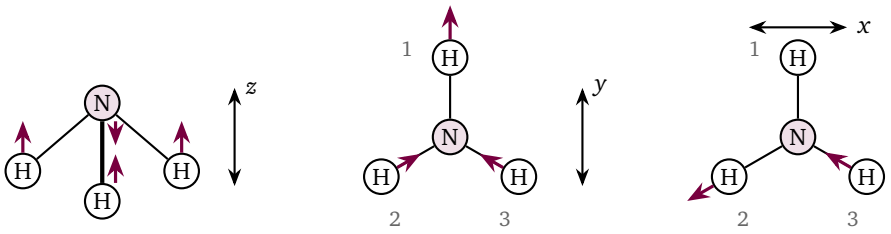
In a normal mode $q(t) = A\, \varepsilon^{(k)} \cos \omega_k t$, the dipole oscillates at ω_k through its linear part, $\vec{d}(q) \approx \vec{d}(0) + Lq$, where L is the $3 \times 3N$ matrix of derivatives $\partial \vec{d}/\partial q$ at $q = 0$. The mode is *infrared active* if $L\varepsilon^{(k)} \neq 0$. Differentiating the display at $q = 0$ gives $L\rho(g) = R(g)L$, so L is an intertwiner from the displacements to $\mathbb{R}^3$, which carries Γ_{vec}. On the isotypic component of type $V^{(i)}$ in V_{vib}, L lies in a space Hom_G that by Lecture 5 is zero unless $V^{(i)}$ occurs in Γ_{vec}:

a mode is infrared active only if its type occurs among the types of x, y, z.

For ammonia, Lecture 5's table gives $\Gamma_{\text{vec}} = A_1 \oplus E$, from z and (x, y). The vibrations are $2A_1 \oplus 2E$ (Lecture 7), so symmetry forbids none of the four fundamentals, and all four are indeed observed in the infrared. The rule can only forbid; whether an allowed line is strong is a question for the dynamics.

Symmetry still fixes the direction in which each mode's dipole oscillates. L maps the A_1 modes into the A_1 part of $\mathbb{R}^3$, the z-axis, and the E modes into the (x, y)-plane. So light polarised along the axis excites only the A_1 modes, and light polarised in the plane only the E modes. Inside one E pair, take partners $\varepsilon_x, \varepsilon_y$ that transform like x and y. On this one copy of E, Schur's lemma gives $L\varepsilon_x = c\,\hat{x}$ and $L\varepsilon_y = c\,\hat{y}$ with a single number c. Light polarised along x excites only the x-like partner, light along y only the y-like one, with equal strength (Figure 8.1). For the E stretch these partners are, up to normalisation and sign, $s_2 - s_3$ and $2s_1 - s_2 - s_3$ of Lecture 7.

In quantum mechanics each normal coordinate Q_k, with $q = \sum_k Q_k \varepsilon^{(k)}$ on V_{vib}, is a harmonic oscillator of frequency ω_k. The vibrational ground state $|0\rangle$ has the wavefunction $\exp\big(-\sum_k \omega_k Q_k^2/2\hbar\big) = \exp\big(-q^{\mathsf{T}} D^{1/2} q/2\hbar\big)$, up to a constant. It is invariant, since $\rho(g)$ is



(a) umbrella (A_1) (b) $2s_1 - s_2 - s_3$ (E) (c) $s_2 - s_3$ (E)

Figure 8.1: Which light excites which mode of ammonia. The coloured arrows are the displacements, schematic as in Lecture 7; the black double arrow is the direction in which the dipole oscillates. (a) An A_1 mode, seen from the side: the dipole oscillates along the axis z. (b), (c) The two partners of the E stretch, seen down the axis with H_1 on the y-axis: the dipole oscillates along y and along x, with the same amplitude.

orthogonal and commutes with D, so $|0\rangle$ is of type A_1. One quantum in mode k multiplies it by a multiple of Q_k, and $\langle 1_k|\, d_\alpha\, |0\rangle = \sqrt{\hbar/2\omega_k}\,(L\varepsilon^{(k)})_\alpha$. So the infrared rule is Theorem 8.8 with $V_T = \Gamma_{\text{vec}}$ and initial type A_1, and the polarisation rule is the Wigner–Eckart theorem: for the aligned partners, $\langle 1_\mu|\, d_\alpha\, |0\rangle = \langle E\|d\|A_1\rangle\, \delta_{\mu\alpha}$ with $\mu, \alpha \in \{x, y\}$.

Raman scattering is the other way a vibration shows up in a spectrum. Light of frequency Ω, far from any absorption line, induces a dipole $\vec{p} = \alpha(q)\,\vec{\mathcal{E}}$, where the *polarisability* $\alpha(q)$ is a symmetric 3×3 matrix. If a mode makes α oscillate at ω_k, the induced dipole oscillates at $\Omega \pm \omega_k$, and the scattered light carries these shifted frequencies: the Raman lines. Under a symmetry operation $\alpha\big(\rho(g)q\big) = R(g)\,\alpha(q)\,R(g)^{\mathsf{T}}$, so the linear part of α is an intertwiner from the displacements to the symmetric matrices. These transform as $S^2\Gamma_{\text{vec}}$ (Proposition 8.5), and so do the quadratic forms $\sum_{a,b} \alpha_{ab} x_a x_b$. The argument above, word for word, gives

a mode is Raman active only if its type occurs among the types of the quadratics $x^2, y^2, z^2, xy, yz, zx$.

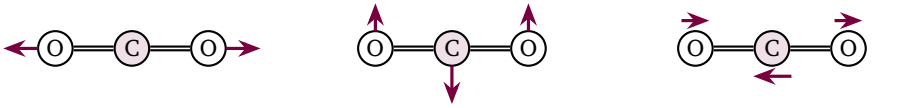
For ammonia $\chi_{\text{vec}} = (3, 0, 1)$ and $\chi_{\text{vec}}(g^2) = (3, 0, 3)$, since the square of a rotation is a rotation and the square of a reflection is E. The proposition gives $\frac{1}{2}\big[(9, 0, 1) + (3, 0, 3)\big] = (6, 0, 2)$, that is $2A_1 \oplus 2E$: z^2 and $x^2 + y^2$ are of type A_1, and $(x^2 - y^2, xy)$ and (xz, yz) of type E. So symmetry allows all four modes of ammonia in the Raman spectrum as well.

Parity and mutual exclusion

In a centrosymmetric molecule the inversion i multiplies all coordinates by -1. So it acts as -1 on Γ_{vec} and as $+1$ on every quadratic. It commutes with every operation of the point group, so by Schur's lemma it acts as $+1$ or -1 on each irreducible representation (Sheet 8, Exercise 5). Each type is therefore even, g (*gerade*), or odd, u (*ungerade*), and so is every mode of that type. An odd mode can only be infrared active, an even mode only Raman active, and no mode is both. This is the *mutual exclusion rule* (Figure 8.2).

8.4 Level splitting under symmetry reduction

Suppose a system with symmetry G is perturbed: an external field is switched on, the atom is placed in a crystal, or an isotope is substituted. After the perturbation only a subgroup



ν_1: sym. stretch (g) IR × Raman ✓ | ν_2: bend (u) IR ✓ Raman × | ν_3: antisym. stretch (u) IR ✓ Raman ×

Figure 8.2: Mutual exclusion in carbon dioxide, a centrosymmetric molecule whose carbon atom sits at the centre of inversion. Each mode is even (g) or odd (u) under the inversion i, and by Section 8.3 it can be infrared active or Raman active, never both. The symmetric stretch is seen only in Raman scattering; the bend and the antisymmetric stretch are seen only in the infrared. The bend is doubly degenerate, since it can take place in any plane through the axis: a linear molecule has $3N - 5 = 4$ vibrations (Lecture 7). Its absorption near 667 cm^{-1} (15 μm) is the band behind the greenhouse effect of carbon dioxide. Only the inversion is used: the full symmetry group of a linear molecule is infinite, since it contains every rotation about the axis.

$H \leq G$ survives as a symmetry. Each G-multiplet $V^{(i)}$ is still an H-representation by restriction. We write it $V^{(i)}{\downarrow}H$, and in general it is a *reducible* one. Its H-irreducible content is read off by the usual character inner product, now taken over H:

$$V^{(i)}{\downarrow}H \;\cong\; \bigoplus_j b_{ij}\, W^{(j)}, \qquad b_{ij} = \left\langle \chi_j^H,\, \chi_i|_H \right\rangle_H ,$$

where the coefficients are called the *branching multiplicities*. Let the level $\mathcal{L}_i$ be isolated and the perturbation H' small. Then the new levels stay close to the old one, and degenerate perturbation theory finds them, to first order, as the eigenvalues of $\pi_i H' \pi_i$ on $\mathcal{L}_i$, where π_i is the orthogonal projection onto $\mathcal{L}_i$. This operator commutes with every $\rho(h)$, $h \in H$, so by Lecture 6's block theorem and its corollary, applied to $V^{(i)}{\downarrow}H$, the level splits into H-multiplets of sizes $\dim W^{(j)}$: generically one for each of the $\sum_j b_{ij}$ copies. Character arithmetic gives the allowed splitting pattern; the sizes of the splittings and the order of the new levels require dynamics.

For example, keep one mirror plane of ammonia and discard the rest. The surviving group is $C_s = \{E, \sigma\}$, with characters $A' = (1, 1)$ and $A'' = (1, -1)$. Restricting the C_{3v} table to these two elements gives $\chi_E|_{C_s} = (2, 0) = A' \oplus A''$: a doubly degenerate E level generically splits into two nondegenerate levels, one symmetric and one antisymmetric under the surviving mirror. The tutorial works this out for a real molecule, and splits the d levels of an atom in a crystal.

[tutorial] **Example 8.10** ($C_{3v} \to C_s$: splitting a doublet)**.** The restriction $C_{3v} \to C_s$ of Section 8.4 has a real molecule behind it (Figure 8.3). In Mulliken's notation the prime and double prime of A', A'' record symmetry and antisymmetry under the one mirror.

Physically, replace one hydrogen of NH_3 by deuterium. The *force field* (the potential) retains the full C_{3v} symmetry, since the forces come from the charges, and an isotope has the same nuclear charge (Lecture 7). But the masses enter the dynamical matrix through the mass-weighted coordinates of Lecture 7, and the masses have only C_s symmetry. Restricted to C_s, ammonia's $\Gamma_{\mathrm{vib}} = 2A_1 \oplus 2E$ becomes $4A' \oplus 2A''$. Nothing in C_s is degenerate, so NH_2D generically has six distinct fundamentals where NH_3 has four. Doubling one mass is not a small perturbation, and the new lines need not stay near their parents. The asymmetric bend of NH_3, at 1627 cm^{-1}, gives lines near 1600 and 1390 cm^{-1} in NH_2D, and the N–D

stretch, near 2506 cm^{-1}, lies far below every N–H stretch. Group theory gives the count and the types, not the positions. ◁

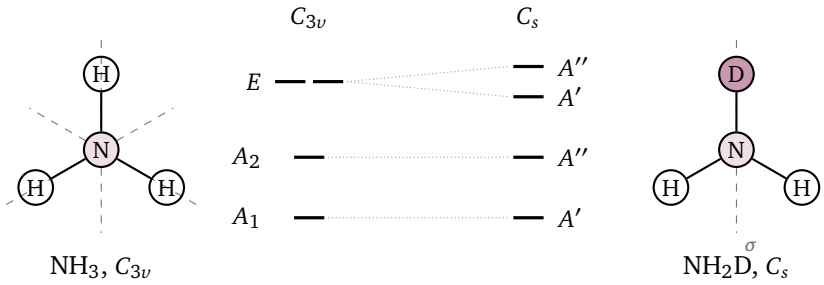


Figure 8.3: Symmetry reduction $C_{3v} \to C_s$, as in Example 8.10. The molecules are seen from above, down the threefold axis of ammonia; dashed lines are the mirror planes, edge-on. Replacing one hydrogen by deuterium keeps only the mirror through the deuterium. In the correlation diagram between them, a level of degeneracy k is drawn as k short bars, and dotted lines follow each level into the smaller group. The doublet E splits into $A' \oplus A''$. Group theory fixes only the connections; the heights of the levels, and the size of the splitting, are schematic.

Example 8.11 (Crystal-field splitting of d levels)**.** A free atom is symmetric under all rotations. Part II will show that its $(2\ell+1)$-fold multiplets are irreducible representations of the rotation group with characters

$$\chi^{(\ell)}\big(C(\theta)\big) = \frac{\sin\big((\ell+\tfrac{1}{2})\theta\big)}{\sin\tfrac{\theta}{2}} \qquad (\theta \neq 0;\ \text{value } 2\ell+1 \text{ at } \theta = 0) \tag{8.1}$$

on a rotation through θ. We borrow this formula now and prove it in Part II.

Place the atom in an octahedral crystal environment: six identical neighbouring ions or molecules, the *ligands*, lie on the coordinate axes. The surviving rotations are those that carry the octahedron to itself. They form the octahedral group O of order 24 (isomorphic to S_4: it permutes the four body diagonals of the cube). The classes of O are E, $8C_3$, $3C_2$, $6C_4$, $6C_2'$, and its character table is

O	$1\,E$	$8\,C_3$	$3\,C_2$	$6\,C_4$	$6\,C_2'$
A_1	1	1	1	1	1
A_2	1	1	1	−1	−1
E	2	−1	2	0	0
T_1	3	0	−1	1	−1
T_2	3	0	−1	−1	1

The full point group $O_h = O \times \{E, i\}$ adds only a parity label, which is even for d orbitals, since they are quadratic in x, y, z. For a d level $\ell = 2$, and formula (8.1) at $\theta = 2\pi/3,\ \pi,\ \pi/2,\ \pi$ gives

$$\chi_d = (5,\ -1,\ 1,\ -1,\ 1),$$

and the inner products, divided by $|O| = 24$, yield

$$m_{A_1} = m_{A_2} = m_{T_1} = 0, \qquad m_E = \tfrac{1}{24}(10+8+6) = 1, \qquad m_{T_2} = \tfrac{1}{24}(15-3+6+6) = 1\ :$$

$$d \longrightarrow E \oplus T_2.$$

The fivefold d level splits into a doublet (e_g, in the chemists' O_h labels) and a triplet (t_{2g}). This is the celebrated crystal-field splitting (Figure 8.4). Its magnitude sets the absorption bands, and hence the colours, of transition-metal complexes. Both new levels are even, and the dipole is odd, so by the selection rule the $d \to d$ line is forbidden in a perfect octahedron. It is seen, weakly, only because vibrations break the inversion for a moment; that is why such colours are pale. The same computation runs unchanged for p and f levels, and, once the improper operations are included, for tetrahedral coordination. It is a worthwhile exercise. ◁

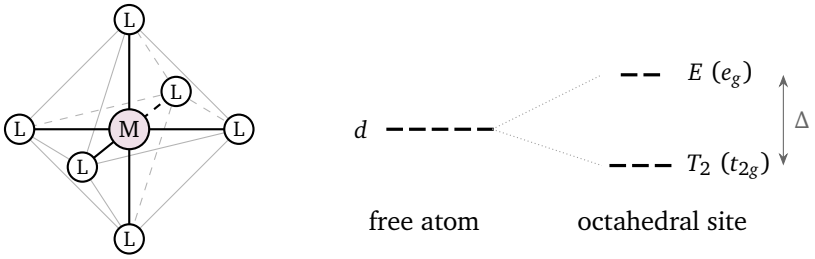


Figure 8.4: Crystal-field splitting. Six identical ligands L on the axes leave the metal atom M the octahedral symmetry, and its fivefold d level splits as $E \oplus T_2$. Group theory decides the pattern $5 = 2 + 3$ and nothing more. That the doublet lies above the triplet for an octahedron, and the size of the gap Δ, come from the dynamics. The single d electron of the titanium ion in the complex $[\mathrm{Ti}(\mathrm{H_2O})_6]^{3+}$ absorbs weakly across this gap near 500 nm, in the green, and the solution looks purple.

8.5 Part I in retrospect *[self-study]*

Eight lectures ago, symmetry was a collection of operations on a triangle. Here is the path since (Figure 8.5). In Lecture 1 we abstracted the operations to groups. In Lecture 2 we mapped out their internal structure: conjugacy, quotients and products. In Lecture 3 groups began to act on vector spaces, and the function-space construction $\psi \mapsto \psi \circ g^{-1}$ was the universal mechanism. In Lecture 4 averaging made every representation unitary and completely reducible, and Schur's lemma turned symmetry into rigid constraints. In Lecture 5 characters compressed all of it into tables. The last three lectures harvested the physics. In Lecture 6 the projectors and the block theorem turned the tables into degeneracies, in Lecture 7 representation theory shaped vibrational spectra, and in Lecture 8 it decided which transitions can be seen and how levels split.

One argument did all the work: *average over the group, and invoke Schur*. One slogan summarises the physics: *what symmetry permits is organised by irreducible representations; what symmetry forbids is exactly zero.* The triangle has carried us as far as it can. From Lecture 9 the groups are continuous, like the rotations of the circle.

Looking ahead to Part II *[self-study]*

Part II asks what survives when the group becomes continuous, that is, when rotations through every angle are allowed, not just multiples of $2\pi/n$. Sums over the group become normalised Haar integrals. For finite-dimensional continuous representations of compact

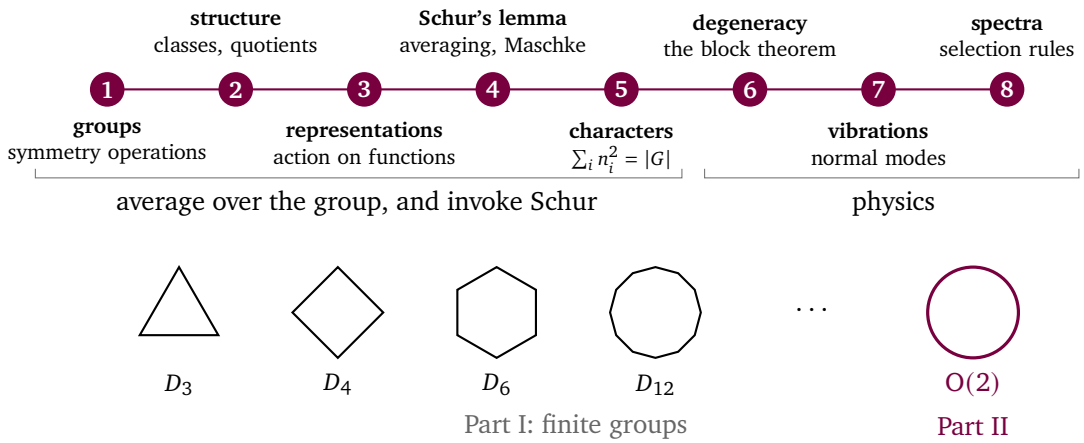


Figure 8.5: Part I in one picture. (a) The eight lectures: five build the machinery and three turn it on physics. One argument carries all of it: average over the group, and invoke Schur. (b) The symmetry groups of regular polygons grow without bound, but every one of them is finite. The circle's symmetry group O(2) contains a rotation through every angle; continuous groups like it are the subject of Part II.

groups, averaging again gives an invariant inner product and complete reducibility. Schur's lemma remains linear algebra. The same averaging argument gives the orthogonality of matrix elements and of characters, and integral projector formulae. Lecture 14 proves these statements for U(1) and SU(2) and develops the corresponding Wigner–Eckart theorem.

The finite counting identities $\sum_i n_i^2 = |G|$ and "number of irreducibles equals number of conjugacy classes" do *not* survive literally. For compact groups the Peter–Weyl theorem replaces them. It states that the matrix elements of irreducible representations are dense in $L^2(G)$, and that the irreducible characters are complete among class functions.

But continuity adds what no finite group possesses: the ability to differentiate. Infinitesimal symmetries form a new algebraic structure, a Lie algebra. Its representation theory yields angular momentum, spin and the spherical harmonics, and with them the proof of the character formula (8.1) borrowed above. Noether's theorem will finally convert symmetry into conservation laws. We develop invariant tensors for SU(2) and leave SU(3) to further study. They prepare the vocabulary for particle-physics applications.

Summary.

- *Tensor products.* $\chi_{V\otimes W} = \chi_V \chi_W$, so products decompose by characters: the Clebsch–Gordan series (Proposition 8.2, Definition 8.3). In C_{3v}, $E \otimes E = A_1 \oplus A_2 \oplus E$: dot product, cross product, and a pair.
- *Selection rule.* A transition caused by an operator of type V_T from a level $V^{(i)}$ to a level $V^{(f)}$ is allowed only if $\langle \chi_f, \chi_T \chi_i \rangle_G \neq 0$ (Theorem 8.8): necessary, never sufficient.
- *Wigner–Eckart theorem.* When $V^{(f)}$ occurs once in $V_T \otimes V^{(i)}$, all the amplitudes are one reduced matrix element times Clebsch–Gordan coefficients (Theorem 8.9).
- *Spectroscopy.* The linear parts of the dipole and of the polarisability are intertwiners. So a mode is infrared active only if its type occurs among x, y, z, and Raman active only if it occurs among the quadratics; with inversion, never both. Symmetry also fixes the polarisation that excites each partner (Wigner–Eckart).
- *Level splitting.* A small perturbation invariant only under a subgroup $H \leq G$ splits an isolated multiplet according to the restriction of its character to H: generically into one

level per H-irreducible copy.

References

Tinkham, *Group Theory and Quantum Mechanics*, §§4-9–4-10, 5-11–5-12 (selection rules and the Wigner–Eckart theorem); Elliott–Dawber, *Symmetry in Physics*, §§4.17–4.20 and Ch. 5; Tung, *Group Theory in Physics*, Ch. 4.